Microbots: Self-Assembling Cubic Micro-Robots Using Electromagnetic Bonding for Reconfigurable Three-Dimensional Structures

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Abstract

Technology has advanced exponentially in the last few decades. We now fabricate chips whose individual transistors are smaller than 10 nm, and many microscopic electromechanical devices already work reliably in everyday products. For years we have imagined micro- and nano-scale robots that move on command and assemble themselves into three-dimensional structures. This paper presents a conceptual design, with preliminary calculations, for such a system, aligned with modular robotics, programmable matter and nanotechnology. The proposed *Microbot* is a cube with a copper coil on each face that acts as an electromagnet, allowing bots to attract, repel and bond with one another. Conductive pads on the edges let bots exchange data and energy, an internal capacitor stores energy, and an on-board controller makes local decisions. The bots carry no battery. Instead, an external oscillating magnetic field induces current in the same coils that are used for bonding, and the bots switch between harvesting and actuation over time. We describe how the design evolved, the architecture, the power scheme, preliminary calculations for coil resistance, heating, magnetic field, bonding force and stored energy, and an updated energy-storage concept based on supercapacitor (porous activated carbon) technology. Current limitations and possible applications are also discussed.

Keywords: microrobots, modular robotics, programmable matter, self-assembly, electromagnetic bonding, wireless power, supercapacitor

1 Introduction

The idea is a theoretical system in which millions of microbots move as commanded and build 3D structures. Each microbot is a small cube that can bond to its neighbours electromagnetically, share data and energy with them, and act as one building block of a larger, reconfigurable structure.

Vision. The motivation behind this project is to build microscopic bots that can move on their own, bond with other bots, and share data and energy with each other. Because each bot is very small, its mass is negligible, but it can still attract or repel other bots and form magnetic bonds. Many bots joined together could form a larger machine with combined computing power.

Definition. *Tessellation*: an arrangement of matter or polygons with no gaps or empty space between them.

Contributions. This paper presents:

- a cubic, face-symmetric microbot design with six electromagnetic coils;
- a combined communication and energy-sharing scheme using edge pads;
- a battery-free power concept based on induction, with time-shared harvesting and actuation;
- preliminary calculations and an updated supercapacitor-based internal storage concept.

2 Design Log

2.1 First design: elongated diamond

The first design was an elongated diamond shape with smooth edges, which looked more like a cylinder. It was discarded for three reasons:

1. the shape allowed at most two joints;
2. organising it into 3D structures was difficult;
3. it was asymmetric, which makes it hard to fabricate at the microscopic scale.

2.2 Requirements for a new shape

To solve these issues, the new structure needed to be:

- symmetric, able to attract and repel other particles at the microscopic level;
- able to pack many particles into a given space (tessellation);
- able to offer the maximum number of joints, so that simple arrangement and de-arrangement algorithms can be used;
- simple to manufacture, symmetric about all three axes, and easy to assemble into 3D structures.

2.3 Final choice: the cube

A cube meets all of these criteria. It tessellates space with no gaps, is symmetric about all three axes, provides six bonding faces, and suits planar lithographic fabrication.

The next challenge was the high resistance of the coil. This was addressed by increasing the coil's cross-sectional area (Section 5).

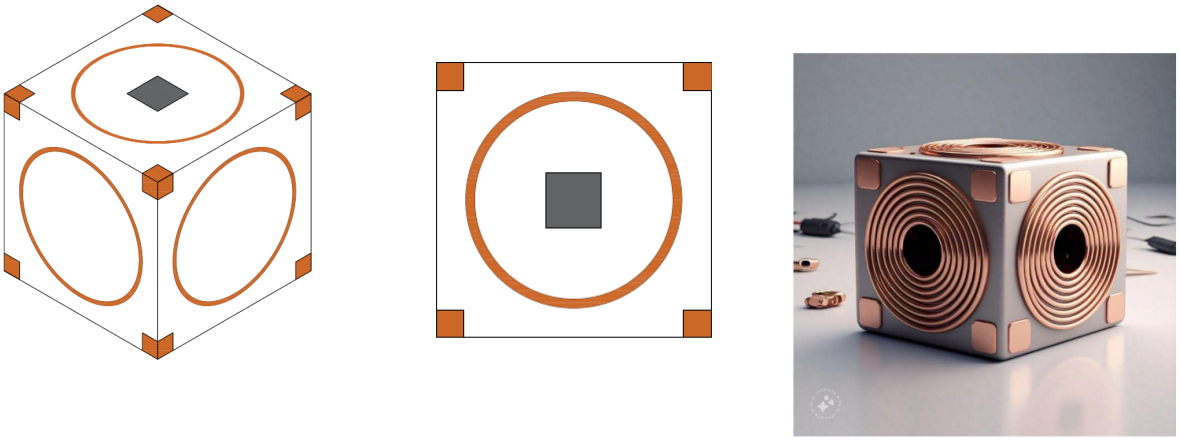
3 Design

Figure 1 shows the design. Each face carries a circular copper coil, a central chip region, and conductive pads at its corners and edges.

3.1 Coils

The bot has a copper coil on each face. The coils can be printed onto the particles using lithography. They serve two purposes:

1. **Magnetic bonding and movement.** Current through a coil produces a magnetic field. When two cubes with opposite polarity come close, they attract each other and form a bond. The coils can also push or pull neighbouring bots, which lets the bots move. The coil current is adjustable, so the push/pull force and the bond strength can be controlled.
2. **Energy harvesting.** Coils that are not in use at a given moment can harvest energy from an external oscillating magnetic field (Section 4).



(a) Isometric view (b) Front (face) view (c) AI-generated render

Figure 1: The Microbot concept: copper coils on each face (orange rings), a central microchip (grey square) and conductive pads (orange squares) on the edges.

3.2 Communication pads

Four conductive (bare copper) pads can be placed on the edges of each face. They handle information transfer between bots and serve two purposes:

1. **Data transfer between bots.** The pads carry commands and data to specific bots, giving the swarm the communication it needs to operate in an organised way. Many bots can also connect together and form a CPU that combines the processing power of every bot.
2. **Energy transfer between bots.** The pads can deliver energy to bots that need more of it. For example, if bots form a sphere, the bots at the centre need much more energy to hold the sphere together than the bots on the outside.

3.3 Internal capacitor

Conductive plates on opposite faces can form an internal capacitor that temporarily stores harvested energy (Figure 2). This storage is useful when:

- several bots are connected at the same time;
- higher instantaneous power is required.

The dielectric or internal material between the plates can be varied, which allows material-based customisation, including PCB-etched conductors or metallic substrates. An updated supercapacitor-based version is described in Section 5.6.

3.4 Internal control (brain)

Each bot can include a miniaturised CPU, microcontroller or logic circuit, fabricated or etched directly onto the bot (for example, a microchip at the centre of each coil). The internal controller enables:

- local decision-making;
- coil activation timing;
- communication with neighbouring bots or an external controller.

Alternatively, many bots can connect together and form a CPU that combines the processing power of each bot.

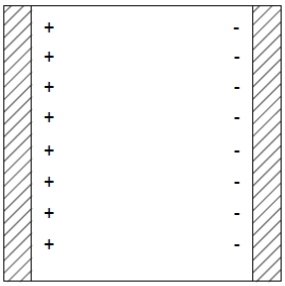


Figure 2: Internal parallel-plate capacitor (older version): plates on opposite faces with a dielectric between them.

3.5 Self-assembly and reconfiguration

Many bots assemble autonomously into predefined or dynamically changing 3D structures based on:

- coil activation sequences;
- external magnetic field parameters;
- local interactions between bots.

Structures can be reconfigured in real time by changing the external field or the activation logic.

3.6 Material

The cube body can be made from materials such as metals or carbon to improve strength and allow variety.

4 Powering the Bots

Powering microbots with miniature batteries is challenging, and it is not necessary here. The microbots have no energy source of their own. Instead, an external oscillating magnetic field induces current in their coils by electromagnetic induction, and the same coils used for bonding also power the bot.

Harvest–use cycle (dual-plexing). At time $t = 0\,\mathrm{s}$ the bot uses its coils to harvest electrical energy, which it stores in its internal capacitor. At $t = 1\,\mathrm{s}$ it uses the stored energy to drive the coils for bonding. The bots repeat this cycle of harvesting and using energy in sync. We call this control technique *dual-plexing*. With it, the bots only need to use their coils for interactions for a few milliseconds each second, so they can recharge their capacitors in between and draw on the stored energy when needed. A full description of the technique is left for future work.

Energy sharing. Some bots may sit in critical positions where they cannot release their magnetic bonds to harvest energy. In those cases, the surrounding bots send energy to them through the communication pads.

As long as an external oscillating magnetic field is present, powering the microbots is not expected to be an issue.

5 Preliminary Calculations

This section reproduces the preliminary hand calculations made for this concept. The calculations cover:

1. the magnetic force between two bots at a certain distance;
2. the current needed to generate that force;
3. the current induced in the coils by the external magnetic field (not yet completed);
4. the heating caused by these currents;
5. the energy that can be stored in the capacitor.

5.1 Coil geometry and resistance

Assume a microbot with a $100\,\mu$m edge. Each face has a ring-shaped coil with an outer diameter of $80\,\mu$m, an inner diameter of $60\,\mu$m and a height of $5\,\mu$m.

Total coil volume:

$$V \approx 11\,000\,\mu\text{m}^3$$

Cross-sectional area $\approx 50\,\mu\text{m}^2$ (wire radius $r = 4\,\mu$m). The length of the equivalent wire is

$$l = \frac{V}{\pi r^2} = \frac{11\,000 \times 10^{-18}}{16 \times 10^{-12}\,\pi} \approx 218.8\,\mu\text{m}$$

and its resistance is

$$R = \frac{\rho\, l}{A} = \frac{1.72 \times 10^{-8} \times 218.8 \times 10^{-6}}{50 \times 10^{-12}} \approx 75.3 \times 10^{-3}\,\Omega.$$

5.2 Heating limit and maximum current

Assuming the cube is made of copper, its mass is about 8.96×10^{-9} kg ($8.96\,\mu$g). Even $34.5\,\mu$J of energy can raise the cube's temperature by $10\,^\circ$C.

Heat produced in the coil is $H = i^2 R t$. With $t = 0.1$ s:

$$34.5 \times 10^{-6} = i^2\,(75.3 \times 10^{-3})(0.1) \;\Rightarrow\; i^2 = 4.58 \times 10^{-3} \;\Rightarrow\; i \approx 62.7\,\text{mA}.$$

So the coils can handle up to 62.7 mA in bursts of 0.1 s each, and this current flowing in one of the cube's six coils raises the bot's temperature by $10\,^\circ$C.

5.3 Magnetic field at the centre of the coil

$$B = \frac{\mu_0 I}{2R} = \frac{4\pi \times 10^{-7} \times 62.7 \times 10^{-3}}{2 \times 35 \times 10^{-6}} \approx 12 \times 10^{-4}\,\text{T}$$

5.4 Force between two bots

The force between two coaxial coils is estimated with

$$F = \frac{3\mu_0 \pi I_1 I_2 R^4 z}{2\,(R^2 + z^2)^{5/2}}.$$

For two bots placed $z = 20\,\mu$m apart, the force of attraction is

$$F \approx 1.25 \times 10^{-10}\,\text{N}.$$

For a bot of mass $8.96\,\mu$g, this corresponds to an acceleration of about $1.395\,\text{m/s}^2$. The force is not bad: it should be enough for the bots to attract or repel each other, although van der Waals forces may also play a role at this scale. Preliminary conclusion: *it can work.*

5.5 Energy stored in a parallel-plate internal capacitor

Assume a bot of size $0.5 \times 0.5 \times 0.5\,\text{mm}^3$, with an internal capacitor roughly the size of the bot (plate side 0.5 mm, separation 0.5 mm). For maximum capacitance, take barium titanate ($BaTiO_3$, $\varepsilon_r = 10\,000$) as the dielectric:

$$C = \frac{\varepsilon_0 \varepsilon_r A}{d} = \frac{8.85 \times 10^{-12} \times 10\,000 \times (0.5 \times 10^{-3})^2}{0.5 \times 10^{-3}} = 44.25 \times 10^{-12}\,\text{F}.$$

The stored energy $E = \frac{1}{2}CV^2$ for different voltages is given in Table 1.

Table 1: Energy stored in the parallel-plate internal capacitor ($C = 44.25\,\text{pF}$).

Voltage V (V)	Energy E
1	22.1 pJ
5	553 pJ
10	2.21 nJ
50	55.3 nJ
100	221 nJ
200	885.4 nJ

Even in the extreme case ($V = 200\,\text{V}$, which is itself difficult to achieve), the capacitor stores only about 885 nJ. That is very little for the bot's operations, so a different design or material is needed.

5.6 Update: supercapacitor-based internal capacitor

Instead of a regular parallel-plate design, the internal capacitor can use the technology behind existing supercapacitors: a porous activated-carbon coating on the electrodes.

Assume a porous carbon coating 100 μm thick (typical values in real devices are 50–100 μm):

$$\text{Volume of carbon} = (0.25\,\text{mm}^2)(0.1\,\text{mm}) = 2.5 \times 10^{-5}\,\text{cm}^3.$$

With a density of $0.5\,\text{g/cm}^3$ (typical values are 0.4–0.6 g/cm^3):

$$\text{Mass of carbon} = (2.5 \times 10^{-5})(0.5) = 12.5\,\mu\text{g}.$$

Real activated-carbon electrodes achieve roughly 100–200 F per gram of material. Taking the conservative value:

$$C = 100\,\text{F/g} \times 1.25 \times 10^{-5}\,\text{g} = 1.25\,\text{mF}.$$

At the typical supercapacitor voltage limit of 2.7 V:

$$E = \tfrac{1}{2}CV^2 = \tfrac{1}{2}(1.25 \times 10^{-3})(2.7)^2 \approx 4.56\,\text{mJ}.$$

Conclusion of the update. The supercapacitor design solves two problems at once:

1. It solves the low-storable-energy problem, raising the stored energy from about 100–1000 nJ to about 4.5 mJ.
2. It removes the need for very high voltages (200 V or more) in a bot whose coils are only about 0.5 mm in diameter. Because the harvesting coils are so small, reaching such voltages would be practically impossible, and sparks would also be a problem at this scale. The coils can realistically generate much lower voltages (about 0–20 V). At those voltages a normal capacitor would store negligible energy, but a supercapacitor, which by nature operates at low voltage (at most about 2.7 V), is a good fit (manufacturing difficulties aside).

As a result, the first item in the limitations list (Section 7.2) can be considered resolved.

6 Applications

The potential applications of this idea are very broad.

1. **Medical applications**
    - *Targeted drug delivery:* delivering medication directly to diseased cells or tissues, which reduces side effects and makes treatment more effective.
    - *Minimally invasive surgery:* performing complex surgery through minimal incisions, reducing patient trauma and recovery time.
    - *Disease diagnosis:* detecting disease at the cellular level, enabling early intervention and personalised treatment.
    - *Biopsy and tissue sampling:* collecting precise tissue samples, improving diagnostic accuracy.
2. **Environmental applications**
    - *Pollution remediation:* cleaning contaminated environments such as oil spills by breaking down or absorbing pollutants.
    - *Environmental monitoring:* measuring air and water quality and providing real-time pollution data.
3. **Industrial applications**
    - *Precision manufacturing:* assembling microscopic components to produce highly complex, miniaturised devices.
    - *Materials science:* manipulating materials at small scales to create new materials with unique properties.
4. **Other potential applications**
    - *Defence and security:* surveillance and reconnaissance.
    - *Space exploration:* exploring and analysing extraterrestrial environments such as planetary or asteroid surfaces.

7 Advantages and Limitations

7.1 Advantages

- The bots act as building blocks, so almost any structure can be made from them.
- The strength of the magnetic bond between two bots can be adjusted.
- Energy is supplied externally, which avoids the problem of on-board batteries.
- Each bot can communicate with other bots and supply energy to them when needed.
- Many bots can join together to form a processor that combines the computing power of each bot.

7.2 Current limitations

1. The internal capacitor cannot store enough energy for the bot *(addressed by the supercapacitor update, Section 5.6)*.
2. The magnetic bond between bots is weak.
3. The energy generated by the coils is very small and not enough to power the bot.
4. Without a proper arrangement algorithm, the bots will end up as a disorganised mass.
5. Manufacturing the bots with high precision can be challenging and costly.

8 Similar Projects

- **M-Blocks** (MIT): self-reconfiguring modular robot cubes that use magnets on their edges and faces to bond and an internal flywheel to move and climb over one another [1].
- **Microscopic autonomous robots (University of Pennsylvania and University of Michigan):** sub-millimetre robots (about $200 \times 300 \times 50\,\mu$m) powered by on-board micro solar cells, with on-board computation and electrokinetic propulsion [2, 3].

9 Future Work

- Complete the calculation of the current induced in the coils by the external magnetic field.
- Publish a detailed description of the dual-plexing control technique.
- Develop arrangement and de-arrangement algorithms for building and reconfiguring structures.
- Improve the bonding force between bots.
- Study fabrication routes (lithography, MEMS processes) for the coils, pads, controller and supercapacitor.

10 Conclusion

This paper presented Microbots: cubic micro-robots that bond, move, communicate and share energy using face-mounted electromagnetic coils and edge-mounted conductive pads, powered wirelessly by an external oscillating magnetic field and buffered by an internal capacitor. Preliminary calculations were presented for coil resistance, heating limits, magnetic field, bonding force and energy storage. Moving from a parallel-plate capacitor to a supercapacitor-style porous-carbon capacitor raises the estimated stored energy from the nanojoule range to about 4.5 mJ at only 2.7 V. The idea is still incomplete and needs much more work and research, but it is shared as a seed for further development. Feedback is welcome at `nitinfromindia4@gmail.com`.

About the Author

Nitin Vishwakarma is from India. He wrote the first version of this concept while in 12th grade (high school) to share the idea and test its feasibility.

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